

GARDEN CITY, KS, USA

WMO: 724515

Lat: 37.927N

Lon: 100.725W

Elev: 878

StdP: 91.21

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23064

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.8	-20.1	0.7	-12.8	-17.6	0.9	-9.3	15.4	5.3	13.9	4.3	4.9	10	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	37.5	20.6	36.0	20.5	34.2	20.4	23.1	31.9	22.5	31.3	21.9	30.7	7.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	17.4	26.4	20.0	16.4	25.5	19.2	15.5	24.9	73.2	31.8	70.5	31.4	68.1	30.7	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	12.4	11.1	DB	-20.5	39.8	2.9	1.5	-22.6	40.8	-24.3	41.7	-25.9	42.6	-28.0	43.7	
			WB	-20.9	24.4	2.8	0.9	-22.9	25.0	-24.6	25.5	-26.1	26.0	-28.2	26.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	0.0	1.8	6.8	11.7	17.6	23.4	26.0	24.9	20.7	13.3	6.0	0.4	(6)
		DBStd	10.58	5.73	6.46	6.08	5.36	4.85	3.81	2.92	3.01	4.46	5.19	5.33	5.54	(7)
		HDD10.0	1193	312	236	132	43	3	0	0	1	26	141	300		(8)
		HDD18.3	2797	569	463	357	205	73	6	1	27	169	369	557		(9)
		CDD10.0	2203	1	6	33	94	239	402	494	461	322	128	21	1	(10)
		CDD18.3	767	0	0	1	7	51	158	237	203	98	13	0	0	(11)
		CDH23.3	9267	0	7	51	196	719	1881	2821	2200	1133	244	14	0	(12)
		CDH26.7	4663	0	1	9	62	304	961	1567	1151	524	84	1	0	(13)
(14)	Wind	WSAvg	5.5	5.1	5.5	6.0	6.5	6.0	5.8	5.0	4.8	5.4	5.5	5.4	5.1	(14)
(15)	Precipitation	PrecAvg	500	9	14	37	43	79	83	75	62	39	24	20	12	(15)
		PrecMax	770	33	47	184	121	202	238	265	176	107	107	93	41	(16)
		PrecMin	262	0	0	0	0	5	2	9	2	2	0	0	0	(17)
		PrecStd	122	9	14	41	33	42	59	55	40	32	25	23	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	20.1	23.9	27.7	31.9	36.1	39.3	39.1	38.9	36.9	32.7	25.3	19.1		(19)
		0.4% MCWB	8.3	10.2	12.8	14.2	17.3	19.1	20.6	20.5	19.1	16.3	11.7	9.4		(20)
		2% DB	16.4	19.7	24.4	28.3	32.8	36.3	37.6	36.5	34.3	28.9	22.2	15.5		(21)
		2% MCWB	6.9	8.5	11.6	13.9	17.4	20.0	21.1	21.1	19.2	15.5	11.0	6.9		(22)
		5% DB	12.7	16.0	21.1	25.5	30.0	34.2	36.2	34.7	32.2	26.0	18.8	12.4		(23)
		5% MCWB	5.1	7.0	10.5	13.0	16.9	20.0	21.2	21.0	19.0	14.7	9.9	5.5		(24)
		10% DB	9.2	12.3	17.7	22.3	27.5	32.3	34.2	32.8	29.8	22.8	15.6	9.4		(25)
		10% MCWB	3.5	5.3	9.4	12.2	16.6	20.1	21.3	20.9	18.6	13.8	8.1	3.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	9.1	11.4	15.7	17.0	20.8	23.4	24.1	23.9	22.1	19.5	14.1	10.3		(27)
		0.4% MCDB	18.2	20.0	22.0	24.0	28.8	31.4	32.7	31.8	30.6	26.1	20.2	16.9		(28)
		2% WB	7.1	9.4	13.5	15.5	19.6	22.4	23.4	22.9	21.1	17.8	12.1	7.6		(29)
		2% MCDB	15.5	18.6	19.9	23.5	27.2	30.9	32.3	31.7	29.7	23.8	18.2	14.7		(30)
		5% WB	5.5	7.6	11.6	14.3	18.7	21.6	22.7	22.2	20.3	16.3	10.7	5.7		(31)
		5% MCDB	12.3	15.3	18.9	22.4	26.3	30.2	31.7	31.1	28.6	22.2	18.0	11.8		(32)
		10% WB	3.8	5.6	9.9	13.0	17.7	20.8	22.0	21.6	19.5	14.9	9.0	4.1		(33)
		10% MCDB	8.9	12.0	17.6	20.7	25.2	29.0	31.0	30.2	27.3	21.6	15.5	9.0		(34)
(35)	Mean Daily Temperature Range	MDBR	14.2	14.5	15.5	15.3	14.7	14.6	14.4	13.8	14.6	15.1	15.0	13.8		(35)
		5% DB MCDBR	20.4	21.1	21.2	20.9	18.8	17.5	17.0	16.8	17.7	19.6	20.3	18.8		(36)
		5% DB MCWBR	12.2	11.7	10.6	9.2	7.1	5.6	5.0	5.2	5.9	8.5	10.8	11.6		(37)
		5% WB MCDBR	19.0	19.7	18.3	17.1	15.7	15.1	14.8	14.6	14.8	14.9	18.0	17.8		(38)
		5% WB MCWBR	11.7	11.3	10.1	8.5	7.1	6.1	5.4	5.4	5.5	7.6	10.0	11.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.268	0.299	0.328	0.348	0.372	0.390	0.372	0.342	0.302	0.271	0.259		(40)
		taud	2.551	2.554	2.469	2.407	2.422	2.392	2.376	2.385	2.454	2.544	2.565	2.559		(41)
		Ebn at Noon	945	978	968	951	931	905	885	895	909	922	923	922		(42)
		Edh at Noon	74	84	101	115	115	119	120	116	102	84	72	68		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.66	3.48	4.74	5.81	6.74	7.38	7.24	6.33	5.32	3.92	2.92	2.35		(44)
		RadStd	0.23	0.35	0.41	0.33	0.41	0.38	0.28	0.34	0.30	0.43	0.25	0.17		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page